The Five-Dimensional Skew-Hermitian Lie Algebra Problem

Automatic Math Discovery Smoke Test

June 14, 2026

Source Problems

The source paper is Kadets–Martin–Paya, *Recent progress and open questions on the numerical index of Banach spaces* [3]. Problem 27 asks for the possible values of $\dim \mathcal{Z}(X)$, where $\mathcal{Z}(X)$ is the Lie algebra of skew-hermitian operators on a finite-dimensional real Banach space X. Problem 28 asks specifically for the five-dimensional case. The source notes that dimensions 3 and 4 were known and that the first open case is dimension 5.

> Our next aim is to discuss some questions related to the Lie algebra of skew-hermitian operators $\mathcal{Z}(X)$ of an arbitrary n-dimensional space which. The main related open question is the following.
>
> **Problem 27.** *Figure out what are the possible values for the dimension of $\mathcal{Z}(X)$ when $\dim(X) = n$.*
>
> Let us fix a n-dimensional Banach space X. It follows from a theorem of Auerbach [72, Theorem 9.5.1], that there exists an inner product $(\cdot|\cdot)$ on X such that every skew-hermitian operator on X remains skew-hermitian (hence skew-symmetric) on $H := (X, (\cdot|\cdot))$. Then, by just fixing an orthonormal basis of H, we get an identification of $\mathcal{Z}(X)$ with a Lie subalgebra of the Lie algebra $\mathcal{A}(n)$. Therefore,
>
> $$\dim\big(\mathcal{Z}(X)\big) \leqslant \frac{n(n-1)}{2}.$$
>
> The equality holds if and only if X is a Hilbert space (see [73, Theorem 3.2] or [60, Corollary 2.7]). It is a good question whether or not all the intermediate numbers are possible values for the dimension of $\mathcal{Z}(X)$. The answer is negative, as a consequence of Theorem 3.2 in Rosenthal's paper [73], which reads as follows.
>
> (a) If $\dim\big(Z(X)\big) > \frac{(n-1)(n-2)}{2}$, then X is a Hilbert space and so $\dim\big(Z(X)\big) = \frac{n(n-1)}{2}$.
> (b) $\dim\big(Z(X)\big) = \frac{(n-1)(n-2)}{2}$ if and only if X is a non-Euclidean absolute sum of $\mathbb{R}$ and a Hilbert space of dimension $n-1$.
>
> For low dimensions, Problem 27 has been solved in [74]. When the dimension of X is 3, the above result leaves only the following possible values for the dimension of $\mathcal{Z}(X)$: 0 as for $X = \ell_\infty^3$, 1 as for $\mathbb{R} \oplus_1 \mathbb{C}$, and 3 as for ℓ_2^3. When the dimension of X is 4, the possible values of the dimension of $\mathcal{Z}(X)$ allowed by the above result are $0, 1, 2, 3, 6$; all of them are possible [74, pp. 443]. The first dimension in which Problem 27 is open is $n = 5$.
>
> **Problem 28.** *What are the possible values for the dimension of $\mathcal{Z}(X)$ when X is a 5-dimensional real Banach space?*

Status of This Packet

This packet is a candidate full solution of Problem 28, the five-dimensional case. It also records a promising but currently conditional route for Problem 27 in all finite dimensions.

The previous promoted draft claimed the all-dimensional route as complete. That was too strong. Its deficit-summand theorem required a nonzero proper G-invariant summand, which is false: for instance $\mathrm{U}(2) \le \mathrm{SO}(4)$ acting on $\mathbb{C}^2_{\mathbb{R}}$ has

$$\dim \mathrm{U}(2) = 4, \qquad w(4) = 5 > 4,$$

but the real representation is irreducible. Likewise $\mathrm{SU}(3) \curvearrowright \mathbb{C}^3_{\mathbb{R}}$ gives a strict triangular deficit without a proper invariant summand. The correct all-dimensional target must allow the subspace W to be all of V, and it must not require W to be G-invariant. What is needed for the Banach-space enlargement argument is a subgroup fixing $W^\perp$ pointwise and acting on W by a proper sphere-transitive subgroup of $\mathrm{SO}(W)$.

Proof Intuition for Problem 28

Auerbach's theorem puts the connected linear isometry group of a finite-dimensional real Banach space inside an orthogonal group, so $\mathcal{Z}(X)$ is a compact Lie subalgebra of $\mathfrak{so}(n)$. Rosenthal's gap theorem says that if $\dim \mathcal{Z}(X) > (n-1)(n-2)/2$, then X is Euclidean. In dimension 5, this leaves only the possible obstruction $\dim \mathcal{Z}(X) = 5$: values $7, 8, 9$ are excluded by Rosenthal, and 10 is the Euclidean case.

Thus the five-dimensional problem reduces to a small compact-Lie-algebra check inside $\mathfrak{so}(5)$. A compact 5-dimensional Lie algebra is either abelian of dimension 5, or has semisimple part $\mathfrak{su}(2)$ and a two-dimensional center. Neither embeds faithfully into $\mathfrak{so}(5)$: an abelian subalgebra of $\mathfrak{so}(5)$ has rank at most 2, and the possible $\mathfrak{su}(2)$-module decompositions in dimension 5 leave centralizer rank at most 1. Hence dimension 5 cannot occur. The remaining displayed values are realized by finite ℓ_∞-sums of Euclidean blocks.

1 The Five-Dimensional Classification

Theorem 1 (Problem 28)**.** *If X is a five-dimensional real Banach space, then*

$$\dim \mathcal{Z}(X) \in \{0, 1, 2, 3, 4, 6, 10\}.$$

Conversely, every value in this set is attained by a five-dimensional real Banach space.

Proof. By the finite-dimensional Auerbach reduction recalled in [3], after choosing an auxiliary Euclidean structure on the underlying real vector space, the identity component of the linear isometry group of X is a compact subgroup of $\mathrm{SO}(5)$, and its Lie algebra is $\mathcal{Z}(X)$. Hence $\mathcal{Z}(X)$ is a compact Lie subalgebra of $\mathfrak{so}(5)$.

Rosenthal's gap theorem for Banach-space isometry algebras [4], as quoted in [3], implies that if

$$\dim \mathcal{Z}(X) > \frac{(5-1)(5-2)}{2} = 6,$$

then X is Euclidean and $\dim \mathcal{Z}(X) = \dim \mathfrak{so}(5) = 10$. Therefore the only value not already eliminated among $0, 1, \ldots, 6, 10$ is 5. It remains to show that no compact 5-dimensional Lie subalgebra of $\mathfrak{so}(5)$ can occur.

Let $\mathfrak{g} \subseteq \mathfrak{so}(5)$ be compact with $\dim \mathfrak{g} = 5$. Its semisimple part is either 0 or $\mathfrak{su}(2)$, since the smallest nonzero compact simple Lie algebra has dimension 3. If $\mathfrak{g}$ is abelian, then it has dimension at most the rank of $\mathfrak{so}(5)$, namely 2, contradiction. Thus

$$\mathfrak{g} \cong \mathfrak{su}(2) \oplus \mathbb{R}^2$$

as a compact Lie algebra.

Consider the underlying faithful orthogonal representation on $\mathbb{R}^5$ and restrict it to the $\mathfrak{su}(2)$-summand. The nontrivial irreducible real $\mathfrak{su}(2)$-modules of dimension at most 5 have dimensions $3, 4, 5$: the 3- and 5-dimensional modules are of real type, and the 4-dimensional module is of quaternionic type.

If the restriction is irreducible of dimension 5, the skew centralizer in $\mathfrak{so}(5)$ is 0, so no two-dimensional center can act faithfully. If the decomposition is $4+1$, the 4-dimensional quaternionic block has orthogonal centralizer $\mathfrak{sp}(1)$, whose maximal abelian subalgebras are one-dimensional, and the trivial line contributes no skew operator. Again a two-dimensional center cannot inject. If a 3-dimensional irreducible block occurs, the remaining fixed part has dimension at most 2, whose skew algebra has rank at most 1; the 3-dimensional real-type block has zero skew centralizer. This also prevents a faithful two-dimensional central action. These cases exhaust dimension 5, so $\dim \mathcal{Z}(X) \neq 5$.

The listed values are realized as follows, where H_k is k-dimensional Euclidean space and the sums are finite ℓ_∞-sums:

$\dim \mathcal{Z}(X)$	X
0	ℓ_∞^5
1	$H_2 \oplus_\infty \ell_\infty^3$
2	$H_2 \oplus_\infty H_2 \oplus_\infty \mathbb{R}$
3	$H_3 \oplus_\infty \ell_\infty^2$
4	$H_3 \oplus_\infty H_2$
6	$H_4 \oplus_\infty \mathbb{R}$
10	H_5.

For such a sum the identity component of the isometry group is the product of the orthogonal groups on the Euclidean blocks, so the Lie algebra dimension is the displayed sum of $\binom{k}{2}$'s. $\square$

2 Triangular Bookkeeping for Problem 27

For $r \geq 2$, put $T_r = \binom{r}{2}$, and define

$$w(N) = \min\left\{\sum_i r_i : N = \sum_i \binom{r_i}{2},\ r_i \geq 2\right\}, \qquad w(0) = 0.$$

Because $T_2 = 1$, $w(N)$ is defined for every $N \geq 0$. Let

$$P(n) = \left\{\sum_i \binom{k_i}{2} : \sum_i k_i = n,\ k_i \geq 1\right\}.$$

Then $N \in P(n)$ if and only if $w(N) \leq n$.

Lemma 2 (elementary support estimates)**.** *The following estimates hold.*

(i) $w(A+B) \leq w(A) + w(B)$.

(ii) $w(\binom{r}{2}) = r$.

(iii) *For* $k \geq 3$, $w(k^2-1) \leq 4k-4$.

(iv) *For* $k \geq 2$, $w(k^2) \leq 2k+1$.

(v) *For $k \geq 1$, $w(k(2k+1)) \leq 2k+1$.*

Proof. Subadditivity follows by concatenating triangular decompositions. The identity $w(\binom{r}{2}) = r$ follows from $\binom{a}{2} + \binom{b}{2} < \binom{a+b}{2}$ for $a, b \geq 2$: with total support $S < r$, the largest triangular sum is at most $\binom{S}{2} < \binom{r}{2}$. For (iii), use

$$k^2 - 1 = \binom{k+1}{2} + \binom{k-1}{2} + (k-2)\binom{2}{2}.$$

For (iv), use $k^2 = \binom{k+1}{2} + \binom{k}{2}$. Finally

$$k(2k+1) = \binom{2k+1}{2}.$$

□

Proposition 3 (realization in every dimension)**.** *Every value in $P(n)$ is attained as $\dim \mathcal{Z}(X)$ for some n-dimensional real Banach space X.*

Proof. If $n = \sum_j k_j$, take

$$X = H_{k_1} \oplus_\infty \cdots \oplus_\infty H_{k_r}.$$

The identity component of the isometry group preserves the finite ℓ_∞-summand decomposition and acts as the product of orthogonal groups on the Euclidean factors. Hence

$$\dim \mathcal{Z}(X) = \sum_j \dim \mathfrak{so}(k_j) = \sum_j \binom{k_j}{2}.$$

□

3 Conditional Route for Problem 27

The expected all-dimensional answer to Problem 27 is:

$$\dim \mathcal{Z}(X) \in P(n) \quad \text{for every } n\text{-dimensional real Banach space } X,$$

with the converse supplied by Proposition 3. The current packet still does not claim this as proved, but the structural flaw in the earlier proof has been repaired: the right object is a defective subspace, not a proper G-invariant summand. The remaining proof-grade work is the compact representation input described below.

Conditional Dependencies

Dependency A: a proof-grade small-module classification. One needs the following representation-theoretic statement, split into machine-checkable pieces rather than a table summary.

- If $\mathfrak{s}$ is compact simple and E is an irreducible faithful real $\mathfrak{s}$-module with $w(\dim \mathfrak{s}) > \dim_{\mathbb{R}} E$, then

  $$(\mathfrak{s}, E) = (\mathfrak{su}(k), \mathbb{C}^k_{\mathbb{R}}) \quad (k \geq 3) \quad \text{or} \quad (\mathfrak{g}_2, \mathbb{R}^7),$$

  up to finite covers, duals, and low-rank isomorphisms.

- If $\mathfrak{s} \oplus \mathbb{R}$ acts irreducibly and effectively with $w(\dim \mathfrak{s} + 1) > \dim_{\mathbb{R}} E$, then

$$(\mathfrak{s} \oplus \mathbb{R}, E) = (\mathfrak{u}(k), \mathbb{C}^k_{\mathbb{R}}),$$

  again up to the standard low-rank identifications.

- If $\mathfrak{h} = \mathfrak{z} \oplus \mathfrak{s}_1 \oplus \cdots \oplus \mathfrak{s}_r$ with $r \geq 2$ acts irreducibly and effectively, then

$$w(\dim \mathfrak{h}) \leq \dim_{\mathbb{R}} E.$$

The supplied supporting file `supporting_small_module_lemma.tex` gives a substantial proof candidate. The Weyl-dimension and product scans described in the verification notes found no suspicious cases. Still, this is the remaining bottleneck: the low-degree tables, real/quaternionic type bookkeeping, and product-factor case should be expanded or cited precisely before Problem 27 is marked full.

Dependency B: compact Lie algebra splitting over semisimple ideals. This part is standard and proof-grade: if

$$0 \longrightarrow \mathfrak{k} \longrightarrow \mathfrak{q} \longrightarrow \mathfrak{a} \longrightarrow 0$$

is an exact sequence of compact Lie algebras and $\mathfrak{s} \subseteq \mathfrak{a}$ is a semisimple ideal, then there is a semisimple ideal $\widetilde{\mathfrak{s}} \subseteq \mathfrak{q}$ mapping isomorphically onto $\mathfrak{s}$, and

$$[\widetilde{\mathfrak{s}}, \mathfrak{k}] = 0.$$

Indeed, a compact Lie algebra is the direct sum of its center and compact simple ideals. The simple ideals of $\mathfrak{q}$ that map nontrivially onto the simple factors of $\mathfrak{s}$ map isomorphically, and the remaining kernel ideals commute with them.

Dependency C: centralizer coupling. The reducible step uses the following centralizer input. Let A be one of the sphere-transitive atoms from Dependency A, acting on a real block E. Let S be its transitive semisimple factor. Suppose S also acts nontrivially on another irreducible real summand

$$F = U \otimes_D D^m, \qquad D = \mathrm{End}_S(U) \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}.$$

If Q acts on $E \oplus F$, projects onto A on E, and has kernel over A contained in the S-centralizer on F, then

$$w(\dim Q) \leq \dim E + \dim_{\mathbb{R}} F.$$

The appendix and the centralizer-coupling interval attempt note give the interval proof. The arithmetic was also stress-tested by the centralizer verifier scripts. The only representation-theoretic reduction used there is the same minimal-module input included in Dependency A.

Corrected Deficit-Subspace Theorem

For $G \leq \mathrm{SO}(V)$ compact and connected, with Lie algebra $\mathfrak{g}$, and for a nonzero orthogonal subspace $W \leq V$, define

$$\mathfrak{g}[W] = \{A \in \mathfrak{g} : \; A|_{W^\perp} = 0\}.$$

Since A is skew-symmetric, $A|_{W^\perp} = 0$ implies $A(W) \subseteq W$.

Definition 4. *The subspace W is* defective *for G if $\mathfrak{g}[W]$ contains a compact connected subalgebra acting transitively on the Euclidean unit sphere of W, but*

$$\mathfrak{so}(W) \not\subseteq \mathfrak{g}[W].$$

Theorem 5 (conditional corrected deficit theorem)**.** *Assume Dependencies A–C. Let $G \leq \mathrm{SO}(V)$ be compact, connected, and effective. If*

$$w(\dim G) > \dim V,$$

then V contains a nonzero defective subspace W.

Proof. Assume not, and choose a counterexample with $\dim V$ minimal. Decompose

$$V = E_1 \oplus \cdots \oplus E_r$$

into irreducible real G-summands. Let H_i be the image of G on E_i, and let K_i be the kernel of the projection $G \to H_i$. At the Lie algebra level,

$$\dim G = \dim K_i + \dim H_i.$$

Suppose $w(\dim H_i) \leq \dim E_i$ for some i. Then K_i acts effectively on $E_i^{\perp}$. If K_i had a defective subspace $W \leq E_i^{\perp}$, then the same W would be defective for G: the transitive subalgebra still fixes the full orthogonal complement of W in V, and if $\mathfrak{so}(W) \subseteq \mathfrak{g}[W]$, those block rotations would act trivially on E_i and hence would already lie in the corresponding kernel algebra. Therefore K_i has no defective subspace on $E_i^{\perp}$. By minimality,

$$w(\dim K_i) \leq \dim E_i^{\perp}.$$

Subadditivity gives

$$w(\dim G) \leq w(\dim H_i) + w(\dim K_i) \leq \dim E_i + \dim E_i^{\perp} = \dim V,$$

a contradiction. Hence every irreducible image satisfies

$$w(\dim H_i) > \dim E_i.$$

By Dependency A, every H_i is one of the strict-deficit sphere-transitive atoms $SU(k)$, $U(k)$, or G_2, up to the listed standard identifications. If $r = 1$, then $W = V$ is defective: the semisimple transitive factor acts transitively on the sphere, while the full $\mathfrak{so}(V)$ is not contained in the proper atom algebra. Thus $r \geq 2$.

Fix $E = E_i$, put $A = H_i$, and let S be the transitive semisimple factor of A. By Dependency B, the S-factor lifts to a semisimple ideal $\widetilde{S} \subseteq \mathfrak{g}$ commuting with the kernel over A. If $\widetilde{S}$ acts trivially on $E^{\perp}$, then E is defective, again because the image of $\mathfrak{g}[E]$ on E lies inside the proper atom algebra $A < \mathrm{SO}(E)$. Hence $\widetilde{S}$ acts nontrivially on some other irreducible summand F.

Let Q be the image of G on $E \oplus F$, and let N be the kernel of the projection $G \to Q$. The projection $Q \to A$ has kernel contained in the $\widetilde{S}$-centralizer on F, so Dependency C gives

$$w(\dim Q) \leq \dim E + \dim F.$$

The kernel N acts effectively on $(E \oplus F)^{\perp}$. As above, a defective subspace for N there would also be defective for G, so minimality gives

$$w(\dim N) \leq \dim(E \oplus F)^{\perp}.$$

Thus

$$w(\dim G) \leq w(\dim Q) + w(\dim N) \leq \dim E + \dim F + \dim(E \oplus F)^{\perp} = \dim V,$$

the final contradiction. $\square$

How the Dependencies Would Prove Problem 27

Assume Dependencies A–C. Let $G = \mathrm{Iso}_0(X)$. By the same Auerbach reduction used above,

$$G \le \mathrm{SO}(V), \qquad \mathrm{Lie}(G) = \mathcal{Z}(X),$$

for an auxiliary Euclidean structure on the underlying real vector space V of X. If

$$w(\dim \mathcal{Z}(X)) > \dim X,$$

Theorem 5 gives a nonzero defective subspace W. For each $u \in W^\perp$, the function

$$x \mapsto \|u + x\|_X \qquad (x \in W)$$

is invariant under a connected subgroup acting transitively on the Euclidean unit sphere of W. Hence this slice norm is radial on W, so the whole norm is invariant under the full $\mathrm{SO}(W)$ acting on W and fixing $W^\perp$. But defectiveness says $\mathfrak{so}(W) \not\subseteq \mathfrak{g}[W]$, so this strictly enlarges the connected isometry algebra, contradicting $G = \mathrm{Iso}_0(X)$. Thus $w(\dim \mathcal{Z}(X)) \le n$, which is equivalent to $\dim \mathcal{Z}(X) \in P(n)$.

Verification Notes

The script `code/check_dimension_values.py` records the triangular sets

$$P(5) = \{0, 1, 2, 3, 4, 6, 10\},$$

$$P(6) = \{0, 1, 2, 3, 4, 6, 7, 10, 15\},$$

and

$$P(7) = \{0, 1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 15, 21\}.$$

These computations verify the finite bookkeeping and construction values, not the conditional all-dimensional representation-theoretic dependencies.

The internal appendix scripts retain the finite and symbolic checks for the centralizer-coupling bounds and the low-dimensional $n = 6, 7$ bookkeeping. The centralizer interval scripts report no failures in the tested ranges, and the Weyl/product verification aids report no suspicious small-module or product-tensor rows. These computations support Dependencies A and C, but they do not replace a written citation or line-by-line proof of the low-degree compact-representation tables in Dependency A.

Novelty Check

Earlier bounded local searches over the parsed corpus and run artifacts found no answer to Problem 28 beyond the source statement. Bounded web searches on June 13, 2026 for exact phrases around $\dim Z(X)$, finite-dimensional real Banach spaces, skew-hermitian operators, and Rosenthal's Lie-algebra paper found adjacent numerical-index and skew-hermitian-operator literature, but no direct later answer to Problems 27 or 28. The Problem 28 result should still be treated as a candidate pending expert review.

Human Review Focus

For Problem 28, review the compact Lie algebra exclusion of $\dim \mathcal{Z}(X) = 5$, especially the centralizer-rank check for the $\mathfrak{su}(2)$-module decompositions.

For the all-dimensional Problem 27 route, do not yet treat this packet as a full proof. The corrected defective-subspace architecture is now included; the main review targets are:

1. expand or precisely cite the low-degree compact-representation table in Dependency A, especially product tensors and real/quaternionic type;
2. check the minimal-module reduction used in the centralizer coupling lemma;
3. verify the corrected defective-subspace induction in Theorem 5;
4. ensure the final Banach-space enlargement is strict through the definition of defectiveness.

References

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